Problem and Scope

1. What specific problem or pain point is your project addressing?

For university students, the long commute time is a big issue. Finding an accessible flat with compatible roommates is the first problem they encounter in university. Our project addresses this issue with 4 major focuses:

1. Compatibility between users
2. Affordability, to minimise cost to users
3. Transparency, to ensure equitable access to information
4. Safety, including Trust and Privacy

2. What research have you conducted to understand the problem better? (Reference supporting studies, statistics, or data collected by your team)

We validated the market needs to understand the problems by performing the following research:

1. Primary User Data: Surveyed 41 student respondents to map specific friction points in finding roommates and verified housing.
2. Macro Market Demand: The Rating and Valuation Department's Private Domestic Rental Index hit a record high of 210.3, driven by heavy seasonal student and talent demand.
3. Institutional Supply Surge: The Development Bureau confirmed that investors are rushing to supply 5,100 student bed spaces through commercial conversions, proving immense market viability.
4. Urgent Security Need: The Hong Kong Police Anti-Deception Coordination Centre has flagged rising rental scams targeting students, highlighting the lack of a secure, trusted marketplace.

Physical supply is growing, but the market critically lacks a centralized, verified platform to safely match these incoming students with suitable housing and roommates.

3. Why is this problem significant? Who does it affect?

Finding suitable roommates and transparency are hard to achieve on current platforms. Hong Kong focuses more on quality education and promotes its education to non-locals in recent years, but there is also a high rent problem in Hong Kong. It affects Hong Kong's reputation and the quality of life of students.

Dividing the problem into two levels:

Individual level: 66% of students are stressed about shared housing expenses, and 68% of students gave up their preferred location due to no roommate.

Structural level: 46% of first-batch Top Talent Pass holders left HK; housing friction is a major factor. There is a 120,000-bed shortfall projected by 2028 because of insufficient resident vacancy in different schools' campuses.

4. What existing solutions are available and how do they fall short?

28Hse, Nestia, Facebook Groups, etc are the current platforms for finding compatible housing and roommates. While each of them has different functions in addressing the problems, they lead to another issue: lack of structured information and verification. For instance:

28Hse offers property listings. It may be a good platform for finding flats, yet it cannot also be used for finding roommates. Plus, it lacks verification and information about the district.

In the current market, there is no platform that integrates housing, roommate matching, and affordable houses with transparent information.

Proposed Solution

5. Describe the primary objectives of your project.

Making the entire process of settling into Hong Kong much smoother for non-local students. This includes matching roommates based on compatibility, finding accurate listings, contract common sense, a multilingual chatbot, and a student community.

6. What are the boundaries of your project (what will the project not address)?

HKID verification, acting on reviews, room tours guided by the platform's employees, dispute settlement, legal advice, services for landlords, payment system, 3D construction of rooms.

7. What assumptions are you making about the problem/ pain point and its context?

Despite halls available for many non-local students, the variety of information scattered online indicates sufficient demand for a structured way to learn about renting in Hong Kong and find a roommate to rent with. Moreover, a university email is sufficient to provide safety and trust for our users. Additionally, users are willing to use the app even with a small user base. Lastly, AI suggestions can add value in aggregating the 'common sense' of Hong Kong, and this is interesting and useful for users, even without full confidence in the information.

8. Who is your target audience or user base?

Nonlocal students who need to settle housing in Hong Kong quickly.

9. Who is your client (the group paying for the solution)?

Universities or higher education institutions in Hong Kong (they may want to connect the service we propose to their school portal so people can find their own places), local and non-local students with housing needs, landlords.

Referral fees from licensed rental property agencies, licensing from universities(to unlock identity verification for their students)

10. How did you gather information about the needs and preferences of your target audience? (Including testing your solution)

Survey of problem statements regarding housing: 41 respondents. Researched housing shortage and competitor markets. Checked HK housing social media forums. Discussed verbally interviewing some stakeholders about their personal experience regarding moving into HK and housing struggles

Teamwork

11. How is your team structured, and what roles does each member play?

Our team is divided between research and design, and technical development. The research team is responsible for understanding user needs, validating the problem, and designing the solution, while the technical team focuses on system architecture, AI integration, and feature feasibility. We work closely together to ensure the solution is both user-centred and technically achievable.

12. What communication strategies are you using to ensure effective collaboration?

We hold weekly online meetings with clear objectives to review progress, track deliverables, and ensure fair distribution of work. This helps keep the team aligned, accountable, and on schedule throughout the project.

Progress

13. What have you achieved so far?

So far, we have validated the problem through primary and secondary research, defined our target audience, and refined our problem statement. We analyzed a 41-response survey to identify key user needs, prioritised core platform features, designed the system architecture, and established the

technical approach for AI-powered matching and verification. These milestones have provided a clear, evidence-based foundation for development.

14. What part of your project are you currently working on?

We are currently developing the prototype and demo while evaluating features based on user value, implementation feasibility, and alignment with our project goals. At the same time, we are refining the prioritisation of MVP features to ensure development efforts focus on the most impactful functionalities.

15. What are your immediate next steps or milestones?

Implement features, for example, an initial onboarding quiz, student verification, identify research API sources, and listing analysis.

16. How will you evaluate the success of your project as it progresses?

First, the initial UI will be generated and observed to confirm the full layout and user flow. The functionality of planned features will be checked on Oct 12 to evaluate completion rated by importance level, and aim for accurate feedback from 5 persons outside this team before Oct 15.

Challenges

17. What challenges or obstacles have you encountered so far? How did you address these challenges?

Due to technical and time limitations, there is a strong need to limit the scope of our project after ideation, for instance, to rule out verification of rental listings for now. More accurately, we cannot gather much feedback since it is difficult to locate our target audience(people who have experienced trying to find rental rooms) among offline audiences or student groups, whether for our initial survey or our prototype. We address the challenges and limitations by finding an alternative feasible solution for features we cannot implement, e.g. using a scraper to locate listings and expecting to cold-call flat owners to upload photos of the rooms with a standard reference object and the floor plan with dimensions.

18. Do you foresee any potential challenges in the upcoming stages of your project? How do you intend to overcome these challenges?

One challenge will be conducting meaningful user testing within the project timeline. To address this, we plan to recruit 5 to 10 international students from universities in Hong Kong for usability testing.

Another challenge will be acquiring sufficient listing data for the prototype. Our approach is to combine manually collected listings with synthetic data to demonstrate platform functionality. We plan to design a landlord listing submission form to support real listing creation and identified partnerships as a potential avenue for expanding listing availability after the hackathon.

Managing AI costs at scale was another important consideration. To minimise expenses, *GPT-4o-mini would be used* only where language generation adds significant value, such as producing compatibility narratives. Core matching and scoring functions would rely on deterministic algorithms, while neighbourhood guide responses would be cached to reduce repeated API usage.

Regulatory compliance presents additional challenges as the platform grows. The prototype is intentionally positioned as a student community tool rather than a licensed property agency. It does not facilitate lease negotiations, process payments, or provide legal advice. If the platform were commercialised and expanded, obtaining the appropriate Estate Agents Authority (EAA) licensing would be explored.

Community moderation is also critical for maintaining platform quality and trust. During the prototype stage, moderation would be handled manually. In a larger-scale implementation, additional safeguards such as automated spam detection, threshold-based content hiding, and verified-student-only posting would help preserve a safe and reliable community environment.

Finally, video calling functionality may be affected by low-bandwidth internet connections. To mitigate this risk, the platform would be connected to existing platforms like Google Meet while also offering alternative viewing methods, such as photo galleries and pre-recorded video walkthrough requests, when live video sessions are not feasible.

Resources

19. What resources (tools, data, libraries, etc.) have you utilised so far in your project? How have these resources contributed to your progress?

Tools include VS Code, WorkBuddy, Microsoft Forms, React, and Supabase. Microsoft Forms allowed primary data collection through surveys; React and Supabase allow us to prototype our design and test our ideas. Workbuddy helps consolidate the information we have found and organise our discussion into tables and notes.

20. Are there any resources (hardware, software, data, expertise) that you need but currently lack?

In the long term, this includes:

- **Real listing data partnership** — using manual + synthetic listings for prototype
- **Land Registry API (IRIS)** — for automated landlord ownership verification; currently users perform land search manually
- **Test cohort of exchange students** — recruiting 5–10 for usability testing
- **Legal review** — formal PDPO/EAA compliance assessment before scaling
- **Granular rental price data** — district-level transaction data for precise price fairness checks
- **Multilingual content validation** — native speakers to verify AI-generated advice in Mandarin, Korean, Japanese, French

Due to these constraints, we will not be able to provide full coverage of data for our users, for now.

Learnings

21. What have you learned so far, either about the problem you're addressing, the technologies you're using, or working as a team?

One of our biggest lessons was that trust is just as important as affordability in shared housing. While students often share accommodation to reduce rental costs, concerns about roommate reliability, lifestyle compatibility, and potential financial disputes strongly influence their housing decisions. This highlighted the importance of building trust mechanisms alongside matching features.

We also learned that AI is most effective when it supports users rather than making decisions for them. Instead of relying entirely on AI-generated recommendations, we used AI to analyse preferences, explain compatibility, and provide insights while letting users make the final decision.

Another key lesson was the value of research-driven development. Our survey of 41 respondents validated demand for verified profiles, compatibility matching, and trust verification, giving us confidence that we were solving a real problem. As a team, we learned that investing time in user research and defining requirements before development reduces uncertainty and helps ensure that the final solution addresses genuine user needs rather than assumptions.

22. How has your understanding of the problem or solution evolved since you started working on it?

We initially viewed the problem as one of affordable housing, assuming students simply needed help finding accommodation within their budget. However, our research revealed that many students can access better housing by sharing rent with roommates, making the real challenge finding people who

are both compatible and trustworthy. Concerns about lifestyle differences, reliability, and potential financial disputes often have a greater impact on the shared living experience than the property itself. This shifted our focus from helping students find accommodation to helping them find roommates they could confidently live with.

As our understanding of the problem evolved, so did our approach to the solution. What began as a simple roommate-finding platform developed into a trust and compatibility platform, incorporating AI-assisted matching, identity verification, and trust-building mechanisms. We also refined our view of technology, recognising that AI should help users make better-informed decisions rather than make decisions for them.

Most importantly, the project reinforced the value of evidence-based design. Our survey of 41 respondents validated key user priorities, including verified profiles, compatibility matching, and trust verification, transforming assumptions into evidence. By completing the empathise phase before development, we ensured that the solution was driven by genuine user needs rather than our own perceptions of the problem.

23. What other issues or comments does your team want to raise?

Regulatory compliance: The platform operates as a student community information tool, not a licensed estate agency. It does not negotiate leases, execute contracts, or handle payments. Verification is PDPO-compliant (purpose-limited, use-limited, clear consent). Building trust within regulatory boundaries is the sustainable path.

AI limitations (per Hack4SDG tips):

- **Room size estimation:** margin of error — presented as a range with confidence, not a precise measurement
- **Scam detection:** false positives and false negatives — presented as "potential concerns," not definitive verdicts
- **Compatibility narrative:** grounded in quiz data; prompts tested to prevent overstating compatibility
- **Multilingual chatbot:** may hallucinate — RAG grounds responses in sourced data; disclaimers included

PPT inclusion list:

Problem statement + data support (survey + statistics) (+ market size? To show demand exists)

Current solution (the gaps)

Our solution
*can include the demo, similar to what peris sent out

Feature inclusion list:

Feature	How It Works
AI Housing Chatbot	"I need a room near HKU, under $5k, quiet" → results + advice
Roommate Matching	Lifestyle quiz → compatibility score + narrative
Listing Scanner	Photo → AI extracts info, flags red flags, compares price
Room Size Estimation	Video/photo → AI estimates real dimensions with confidence scores
Virtual Visit	3D model or photo-based room analysis for remote students
Verified Community	University email verification + reviews

USER JOURNEY:
REGISTER / SIGN IN
VERIFICATION (upload student identity card + uni email)
FILL IN FORM (lifestyle + budget + preferences)
GET COMPATIBILITY SCORES WITH EXISTING PEOPLE(Ranking in accordance with compatibility score, with the highest percentage): THE COMPATIBILITY IS BASED ON LIVING HABITS AND THEIR BUDGET
They can move on to communicating on the platform chat feature
After communicating, they can house hunt on the platform listing and rank their housing preferences
Reach out to the landlord for negotiation
Sign a living agreement
GET AFFORDABLE GOOD HOUSING!